

Pramukh Kashyap R M

B.Tech (5th Sem) – Computer Science and Engineering | Sardar Vallabhbhai National Institute of Technology
pramukhkashyaprm@gmail.com | +91 9150496652 | GitHub | LinkedIn

PROFILE

Computer Science undergraduate with strong foundations in DSA, DBMS, OS, and software engineering. Experienced in building backend systems, AI applications, and simulations. Interested in backend engineering, AI/ML, and quantitative software development.

EDUCATION

Qualification	Institution	Year	Performance
B.Tech in Computer Science	SVNIT, Surat	2024 – 2028 (Expected)	CGPA: 8.37
Class XII	CBSE	2024	93%
Class X	CBSE	2022	96%

PROJECTS

Traffic Simulation System: Discrete-Time System *C++, SFML, Object-Oriented Design*

- Built an OOP real-time traffic simulation engine modeling urban road topologies, dynamic vehicle acceleration, and bottlenecks.
- Implemented A* route planning, graph-based shortest path algorithms, and adaptive signal control to reduce congestion.

FinSight: Hybrid Financial Retrieval Backend *Python, FastAPI, PostgreSQL, pgvector, SQLAlchemy*

- Engineered an asynchronous search engine decoupling qualitative text search from quantitative financial metrics to prevent hallucinations.
- Built a dual-stage pipeline with FastAPI routing semantic queries to **pgvector** and structured data to PostgreSQL.
- Optimized vector indexing via HNSW graph migrations to accelerate high-dimensional approximate nearest-neighbor search.

LogiSwarm: Centralized Multi-Agent RL Navigation Framework *Python, ROS2, Gazebo, Stable-Baselines3, PyGame, NumPy*

- Developed a multi-agent pathfinding and collision-avoidance framework using ROS2, Gazebo, and Stable-Baselines3.
- Prototyped a 2D grid-world in PyGame/NumPy for custom reward formulation before deployment to 3D Gazebo.
- Trained centralized multi-agent RL policies on Linux with GPU acceleration, validating path planning with real-time visualization.

ACHIEVEMENTS

- Finalist, Genesis Hackathon:** Selected among top teams for designing an ML-enhanced Algorithmic Trading & Backtesting Framework.

TECHNICAL SKILLS

Programming: C++, C, Python, SQL, Assembly (8085/8086)
Core CS: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks
Frameworks & Libraries: SFML, FastAPI, SQLAlchemy, ROS2, Stable-Baselines3, NumPy, Pandas, Scikit-learn
Databases & Tools: PostgreSQL, pgvector, Gazebo, Git, GitHub, Linux (Ubuntu)

RELEVANT COURSEWORK

- Data Structures & Algorithms | Design and Analysis of Algorithms | Computer Architecture | Object-Oriented Programming
- Operating Systems | Database Management Systems | Computer Networks | Artificial Intelligence

DOMAIN FOCUS & TECHNICAL INTERESTS

- Backend Engineering:** REST API development, asynchronous programming, database design, and scalable architectures.
- AI & Data-Driven Systems:** Retrieval-Augmented Generation (RAG), machine learning, vector databases, and information retrieval.
- Simulation & Quantitative Computing:** Algorithm design, optimization, reinforcement learning, and simulation modeling.